

SESSION 17 — JOB CONTROL IN UBUNTU

Task 1: List Active Jobs

Question

To implement how to list active jobs.

```
ganesh@DESKTOP-79N3772: ~  
ganesh@DESKTOP-79N3772:~$ sleep 300 &  
[1] 521  
ganesh@DESKTOP-79N3772:~$ sleep 400 &  
[2] 538  
ganesh@DESKTOP-79N3772:~$
```

To display status information of active jobs.

```
ganesh@DESKTOP-79N3772:~$ jobs -l  
1)- 521 Running sleep 300 &  
2)+ 538 Running sleep 400 &  
ganesh@DESKTOP-79N3772:~$
```

To retrieve the Job IDs of active jobs.

```
ganesh@DESKTOP-79N3772:~$ jobs -p  
521  
538  
ganesh@DESKTOP-79N3772:~$ jobs -x echo %1  
521  
ganesh@DESKTOP-79N3772:~$
```

```

ganesh@DESKTOP-79N3772:~$ sleep 300 &
[1] 521
ganesh@DESKTOP-79N3772:~$ sleep 400 &
[2] 538
ganesh@DESKTOP-79N3772:~$ jobs -l
[1]- 521 Running                  sleep 300 &
[2]+ 538 Running                  sleep 400 &
ganesh@DESKTOP-79N3772:~$ jobs -p
521
538
ganesh@DESKTOP-79N3772:~$ jobs -x echo %1
521
ganesh@DESKTOP-79N3772:~$ jobs -p
521
538
ganesh@DESKTOP-79N3772:~$ kill -STOP %1

[1]+  Stopped                  sleep 300
ganesh@DESKTOP-79N3772:~$ jobs
[1]+  Stopped                  sleep 300
[2]-  Running                  sleep 400 &
ganesh@DESKTOP-79N3772:~$ bg %1
[1]+ sleep 300 &
ganesh@DESKTOP-79N3772:~$ fg %1
sleep 300
Ctrl + Z

```

```

ganesh@DESKTOP-79N3772:~$ kill %1 %2
-bash: kill: %1: no such job
[2]+  Terminated              sleep 400
ganesh@DESKTOP-79N3772:~$ jobs

```

Observation:

The practical demonstrates Linux job-control commands such as `jobs`, `fg`, `bg`, `wait`, and `kill` to list, monitor, stop, resume, switch, and control foreground and background processes.

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TASK 1

Question: To analyze signal delivery, understand signal masks, review process groups, explore terminal signals, and test signal behavior.

```

ganesh@DESKTOP-79N3772:~$ sleep 300 &
[1] 799
ganesh@DESKTOP-79N3772:~$ ps
  PID TTY          TIME CMD
  332 pts/0    00:00:00 bash
  799 pts/0    00:00:00 sleep
  809 pts/0    00:00:00 ps
ganesh@DESKTOP-79N3772:~$ ps -o pid,ppid,pgid,sid,stat,cmd -p $!
  PID   PPID    PGID     SID STAT CMD
  799    332     799     332 S   sleep 300
ganesh@DESKTOP-79N3772:~$ kill -STOP $!

[1]+  Stopped                  sleep 300
ganesh@DESKTOP-79N3772:~$ ps -o pid,stat,cmd -p $!
  PID STAT CMD
  799 T   sleep 300
ganesh@DESKTOP-79N3772:~$ kill -CONT $!
ganesh@DESKTOP-79N3772:~$ ps -o pid,stat,cmd -p $!
  PID STAT CMD
  799 S   sleep 300
ganesh@DESKTOP-79N3772:~$ kill $!
[1]+  Terminated              sleep 300
ganesh@DESKTOP-79N3772:~$ ps -p $!
  PID TTY          TIME CMD
ganesh@DESKTOP-79N3772:~$ _

```

Observation: Signals such as `SIGSTOP`, `SIGCONT`, and `SIGTERM` are used to control and terminate processes.

TASK 2

Question: To resume stopped jobs, send continue signals, update job status, maintain job table, verify execution, and test background recovery.

```

ganesh@DESKTOP-79N3772:~$ sleep 300 &
[1] 902
ganesh@DESKTOP-79N3772:~$
ganesh@DESKTOP-79N3772:~$ jobs -l
[1]+  902 Running                  sleep 300 &
ganesh@DESKTOP-79N3772:~$ kill -STOP %1

[1]+  Stopped                      sleep 300
ganesh@DESKTOP-79N3772:~$ jobs
[1]+  Stopped                      sleep 300
ganesh@DESKTOP-79N3772:~$ kill -CONT %1
ganesh@DESKTOP-79N3772:~$ jobs
[1]+  Running                      sleep 300 &
ganesh@DESKTOP-79N3772:~$ fg %1
sleep 300
^Z
[1]+  Stopped                      sleep 300
ganesh@DESKTOP-79N3772:~$ bg %1
[1]+  sleep 300 &
ganesh@DESKTOP-79N3772:~$ jobs -l
[1]+  902 Running                  sleep 300 &
ganesh@DESKTOP-79N3772:~$ jobs
[1]+  Running                      sleep 300 &
ganesh@DESKTOP-79N3772:~$

```

Observation: Stopped background jobs can be resumed using `SIGCONT`, and their execution status can be monitored using the job table commands.